GENAI BASICS • LESSON 08 • MODULE 3: AI ASSISTANTS AT WORK

AI for Data Analysis

Spreadsheet → answer → chart → one-page report

Formulas, pivots and charts on request in plain words — in Excel, Google Sheets or a chat with a file. Case study: Northstar online shop sales for 2025.



WHAT THIS LESSON GIVES YOU

Five skills you take away today

1. **Describe a file**
   understand what is in a table in a minute
2. **A formula in words**
   ask for a formula and a pivot in Excel
3. **A slide-ready chart**
   the type fits the question, the title states the finding
4. **A one-page report**
   a file you can forward
5. **A one-minute check**
   rows, one point, percent of what

THE TASK WE LEARN ON

An e-mail from the commercial director: “How did sales go this year? Why did advertising pay off worse in the second half? And why so many returns in the autumn? I need one page by Friday.”

> We spend the lesson answering this e-mail. At the end we have the report page.

Where data lives at work



Spreadsheets

Excel, Google Sheets: sales, budget, plan

`AI writes formulas and pivots`



Exports

CSV from a CRM, accounting, ad accounts, YouTube Studio

`file into a chat, AI writes code`



Databases

CRM, website, warehouse: millions of rows, updated daily

`AI writes an SQL query`



Text

reviews, e-mails, support chats, surveys

`AI reads and sorts into themes`



Documents

PDF invoices, contracts, partner reports

`AI pulls the numbers into a table`



Events and logs

clicks, orders by the minute, calls

`usually an analyst queries the database`

Most often you have a spreadsheet or an export. Sometimes text. Less often a database. Today we cover all three.

Table and text: AI handles them differently

Sales sheet: numbers, dates, categories

date	channel	category	region	orders	revenue
2025-01-01	Social	Home	Central	10	580
2025-01-02	Organic	Sports	West	11	670
2025-01-02	Social	Books	West	18	400
2025-01-02	Social	Home	West	16	890
2025-01-03	Organic	Electronics	South	39	3,660
2025-01-03	Organic	Sports	North	19	1,170

AI counts: it writes a formula or code, a program adds and divides. The typical error is a wrong formula.

Reviews sheet: customers' own words

★★★★★ All great, arrived fast. Thanks.
C1 · 2025-01-04

★★★★ The product matches the description. Will order again.
N1 · 2025-01-06

★ Badly packed — no wrap, no foam, scratched all over.
N2 · 2025-09-07

★★ Courier handed over a wet, torn box. Thanks.
N2 · 2025-09-07

AI reads: it assigns each review to a theme and counts themes. The typical error is the wrong theme.

One row: what the record is and what it measures

order_date	channel	category	region	warehouse	orders	revenue	ad_spend	returns
2025-01-05	Email	Electronics	North	N2	11	980.0	0	1
date	category	category	category	category	number, pcs	number, $	number, $	number, pcs

A row is the sales of one channel for one category, region and warehouse on one day. Not an order, not a customer.

Categories are what we slice by: channel, region, warehouse. Breakdowns are built on them.

Numbers are what we add and divide. Each has its unit: dollars or pieces.

First question for any file: what is one row? Get it wrong and every “average per customer” is wrong.

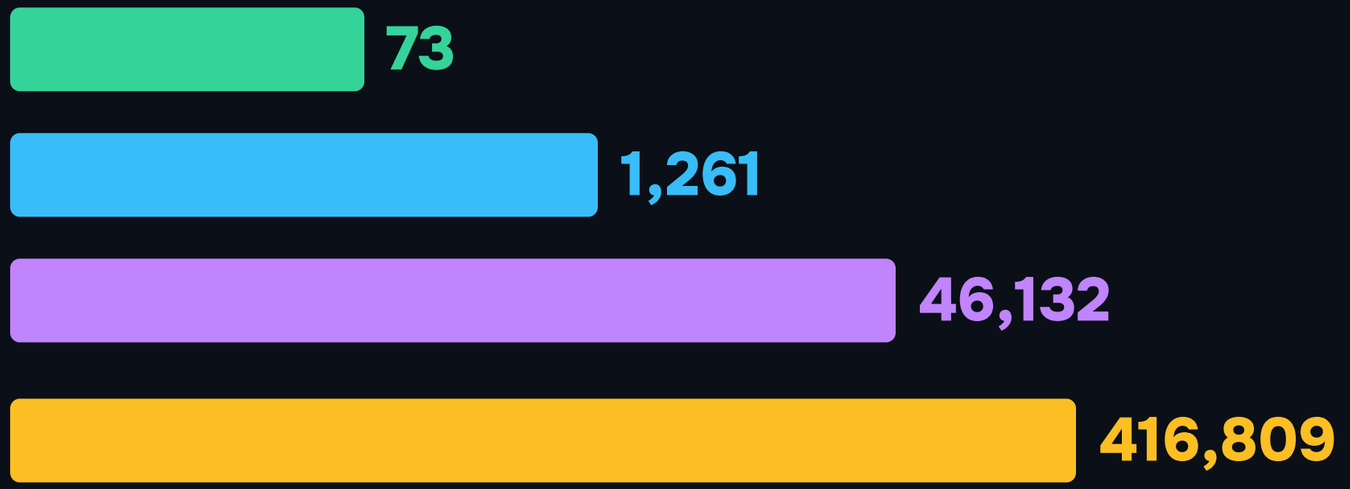
Size decides where the data goes

dozens of rows
can be pasted into a message

thousands of rows
our Northstar file: upload as a file

tens of thousands
channel DB sample: a file still works

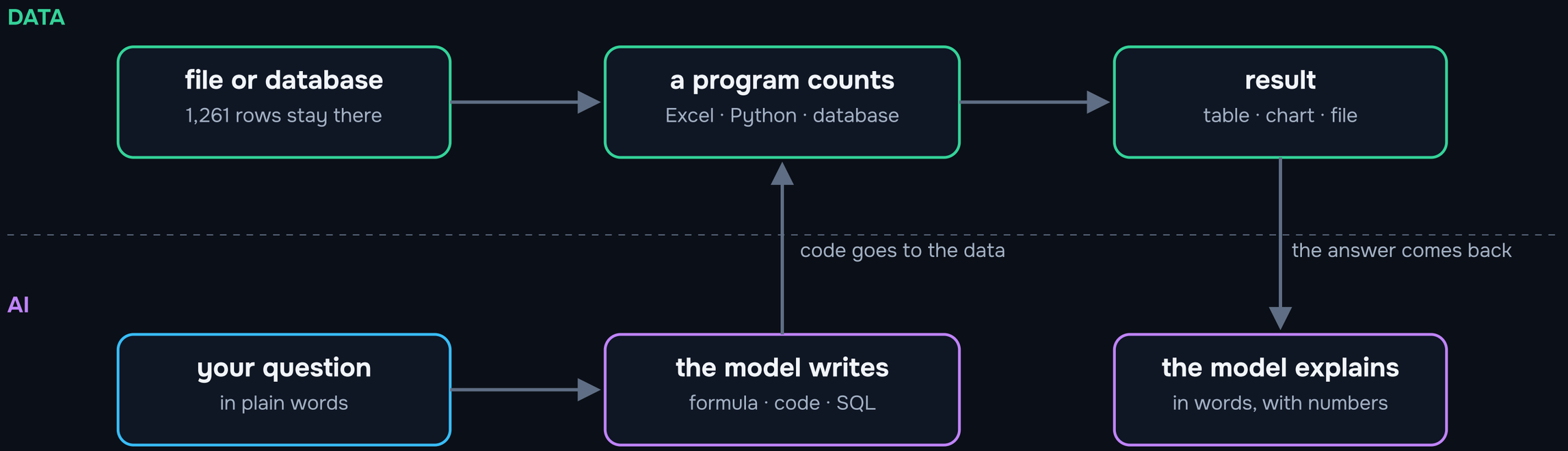
hundreds of thousands+
the whole video DB: ask the database with a query



Log scale: each step is roughly 10× the previous one.

PART 1 • THE KEY DIAGRAM

Data stays in the file. AI writes instructions, a program counts



You stand at two points: asking the question and checking the result.

Rule: when AI states a number, ask which program computed it.

PART 1 • REAL DATA

Four traps in real data, three of them in our file

A duplicated row

`2025-03-19 · Social · Home · 3,890.5 ×2`

revenue is inflated by this amount

A date as text

`10/05/2025 ← 2025-05-10?`

10 May or 5 October? Depends on the format

Zero or missing?

`Email · ad_spend = 0`

Email has no ads. The model called these zeros “missing”

Mixed units

`1,200 (RUB) and 15 ($) in one column`

a real-life example: they can’t be added

The model often finds these, but doesn’t always understand them. The data owner knows what a zero means and which date is right.

73 rows pasted into a message: the model got it right. Why we still don’t do it

PROMPT: 73 CSV ROWS, “NO TOOLS OR CODE”

```
1001,2026-01-04,Search,Electronics,North,36,1656.0,306.08,1
1002,2026-01-13,Search,Home,South,41,2132.0,353.76,2
1003,2026-01-22,Search,Sports,West,46,2944.0,515.92,4
1004,2026-01-04,Social,Home,South,43,2236.0,599.36,3
1005,2026-01-13,Social,Sports,West,46,2898.0,776.48,3
1006,2026-01-22,Social,Books,Central,54,3834.0,967.84,5
1007,2026-01-04,Email,Sports,West,42,2520.0,0,2
1008,2026-01-13,Email,Books,Central,50,3550.0,0,4
1009,2026-01-22,Email,Electronics,North,54,2376.0,0,2
1010,2026-01-04,Organic,Books,,44,2992.0,0,4
1011,2026-01-13,Organic,Electronics,North,53,2491.0,0,1
1012,2026-01-22,Organic,Home,South,59,3068.0,0,3
1013,2026-02-04,Search,Home,West,40,2140.0,408.2,2
1014,2026-02-13,Search,Sports,Central,48,3000.0,577.0,2
...
```

Показать ещё

CHATGPT ANSWER, 17 OCT 2026

259 357,00 — сумма `revenue` по всем **73 строкам** CSV.

256 045,00 — после исключения одного полного дубля, **72 строки**. Валюта не указана.

2) Проблемы в данных

`order_id`	Проблема
1028	Строка записана **дважды**. Завышение выручки на **3 312,00**.
1010	Не заполнен `region`.
1041	Дата `2026/04/22` отличается форматом от остальных: используются `/` вместо `-`.
1055	**66 возвратов при 62 заказах.** Требует проверки: возможна ошибка либо возвраты относятся к заказам другого периода.

The total is right; the duplicate and blank region found.

No visible working. Nothing to repeat or check.

1,261 rows won’t paste like this. You need a file and code.

Three questions in the e-mail — three parts of the lesson

1. **How did sales go this year?**
   Part 2: Excel, formulas, seasonality
2. **Why does advertising pay off worse?**
   Parts 2–3: Search and Social return
3. **Why the autumn returns?**
   Part 3: the table + review text

NORTHSTAR_2025.XLSX • 1,261 ROWS • REVIEWS SHEET: 357 REVIEWS

A1 fx order_date

	A	B	C	D	E	F	G	H	I
1	order_date	channel	category	region	warehouse	orders	revenue	ad_spend	returns
2	2025-01-01	Social	Home	Central	C1	10	580.00	135.10	0
3	2025-01-02	Organic	Sports	West	W1	11	670.00	0.00	0
4	2025-01-02	Social	Books	West	W1	18	400.00	103.34	1
5	2025-01-02	Social	Home	West	W1	16	890.00	222.15	1
6	2025-01-03	Organic	Electronics	South	S1	39	3,660.00	0.00	2
7	2025-01-03	Organic	Sports	North	N1	19	1,170.50	0.00	1
8	2025-01-03	Search	Electronics	Central	C1	36	3,500.00	626.28	2
9	2025-01-04	Email	Home	North	N1	24	1,210.50	0.00	1
10	2025-01-04	Organic	Home	West	W1	11	610.50	0.00	1
11	2025-01-04	Social	Books	West	W1	22	490.50	117.76	1
12	2025-01-04	Social	Electronics	North	N2	35	3,250.50	795.72	1
13	2025-01-05	Email	Electronics	North	N2	11	980.00	0.00	1
14	2025-01-05	Search	Home	Central	C1	47	2,480.50	449.10	2
15	2025-01-05	Search	Home	South	S1	68	3,440.50	632.18	3
16	2025-01-05	Social	Home	West	W1	18	1,030.50	254.28	1
17	2025-01-06	Organic	Electronics	West	W1	9	840.50	0.00	0
18	2025-01-06	Organic	Home	West	W1	17	970.50	0.00	1
19	2025-01-07	Email	Electronics	Central	C1	28	2,440.50	0.00	1
20	2025-01-07	Organic	Books	Central	C1	54	1,340.00	0.00	3
21	2025-01-08	Email	Sports	Central	C1	23	1,390.50	0.00	1
22	2025-01-08	Email	Sports	South	S1	16	920.00	0.00	1
23	2025-01-08	Email	Sports	West	W1	13	820.00	0.00	1
24	2025-01-08	Search	Home	West	W1	30	1,610.00	300.21	1
25	2025-01-09	Email	Electronics	South	S1	30	2,810.50	0.00	2
26	2025-01-09	Search	Books	South	S1	31	730.50	142.49	1
27	2025-01-09	Search	Sports	Central	C1	18	1,080.00	209.35	1

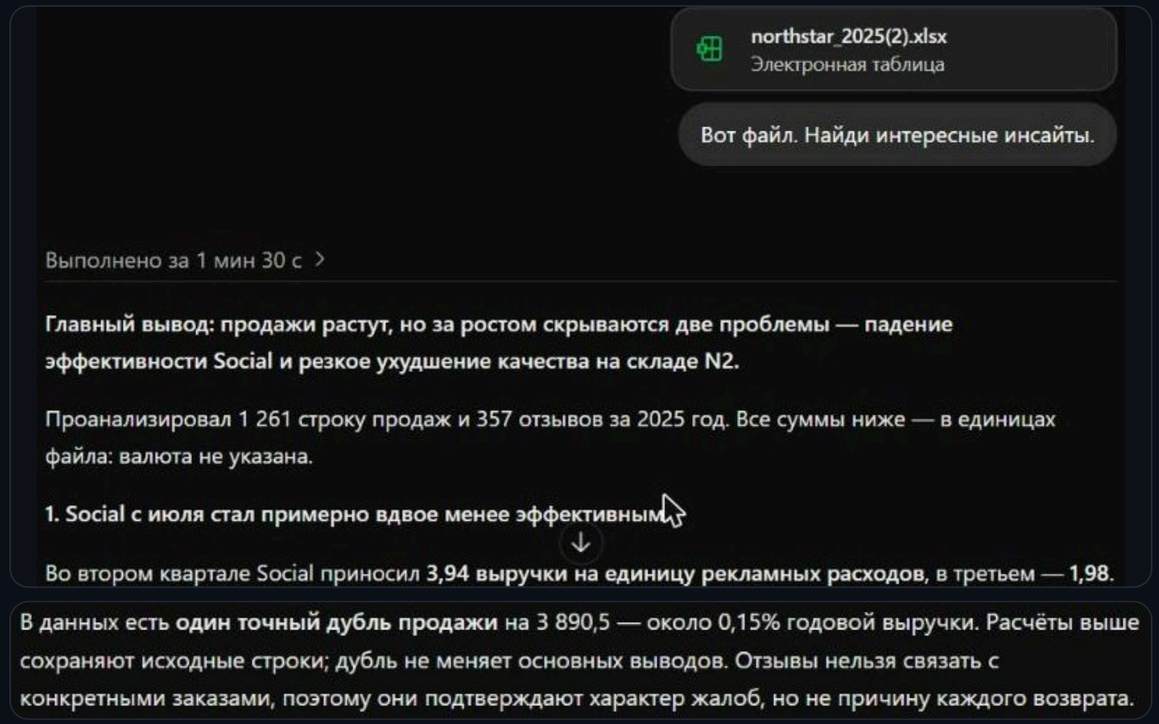
Sales Reviews Ad ROAS 2025

The first prompt for any file: a description

1	`Describe this file: what one row is,`	understand the data first, not compute
2	`how many rows, what period,`	numbers you'll check later
3	`what data problems.`	duplicates and oddities — before any maths
4	`Suggest 5 questions a manager would ask.`	a work plan: pick the question from the e-mail
5	`Don't change anything yet.`	or the model quietly "cleans" the file

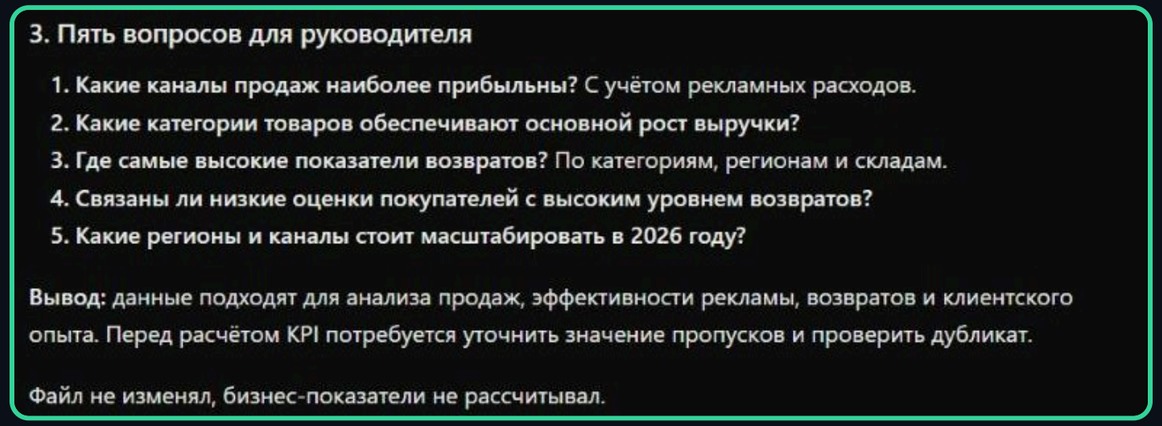
“Find insights” works too. But the model picks the questions

QUICK “Here is the file. Find interesting insights.”



> The model found Social and N2. But it chose the comparisons itself (quarters) and **kept the duplicate in the maths**.

✓ UNDER CONTROL Prompt 1: description + 5 questions + change nothing



> ✓ 1,261 rows, period, duplicate · ✗ called 632 zero ad-spend values “missing” · ✗ missed the text date. **We take question 1.**

A formula in words: each line has a job

1	`Add a sheet “Ad return” to the workbook:`	where the result appears
2	`by month … for Search and Social.`	the breakdown: time and channels
3	`revenue, ad spend and revenue per $1 of ads.`	the metric: revenue ÷ spend
4	`Use Excel formulas, not pasted numbers.`	so you can check and recalculate
5	`Build a line chart`	the type for “how did it change”
6	`and let me download the .xlsx.`	the result is a file, not chat text

Numbers in cells can’t be checked. Formulas can

QUICK Sheet from ChatGPT: plain numbers in the cells

Where 5.39 comes from is invisible. Change the file — numbers don’t update.

A1 fx Month

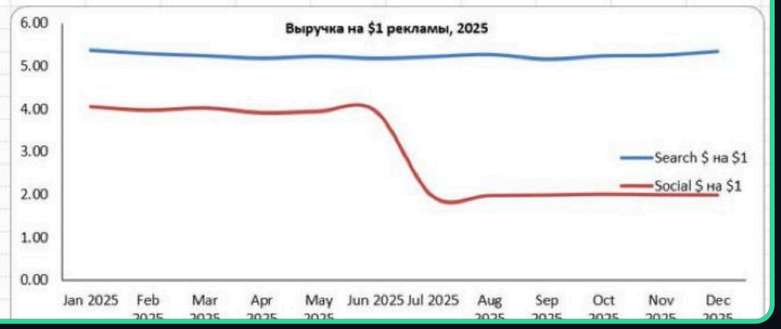
	A	B	C	D	E	F
1	Month	Search revenue	Search ad spend	Search ROAS	Social revenue	Social ad spend
2	Jan 2025	44,755.00	8,310.23	5.39x	38,985.00	9,614.32
3	Feb 2025	45,949.00	8,676.11	5.30x	42,677.00	10,746.11
4	Mar 2025	54,696.50	10,439.37	5.24x	50,378.00	12,514.00
5	Apr 2025	56,835.00	10,977.28	5.18x	41,104.50	10,501.66
6	May 2025	64,157.50	12,283.61	5.22x	54,057.50	13,700.63
7	Jun 2025	58,377.00	11,281.95	5.17x	48,726.00	12,325.24
8	Jul 2025	50,537.50	9,689.36	5.22x	43,154.00	21,948.53
9	Aug 2025	50,739.00	9,620.82	5.27x	49,856.00	25,111.57
10	Sep 2025	61,998.50	12,027.74	5.15x	55,541.00	27,804.92
11	Oct 2025	69,634.00	13,300.06	5.24x	54,668.50	27,070.16
12	Nov 2025	100,933.00	19,217.13	5.25x	89,567.50	44,686.85
13	Dec 2025	117,556.50	21,956.98	5.35x	100,727.00	50,387.44
14						
15	ROAS = revenue / ad spend; revenue is attributed to channel, not proven incremental.					

✓ UNDER CONTROL Excel: a SUMIFS formula in the fx bar

You can see which columns are summed for which month. The fix is one phrase: “replace the numbers with formulas”.

B2 fx =SUMIFS(Sales!$G:$G,Sales!$B:$B,"Search",Sales!$A:$A,">="&$A2,Sales!$A:$A,"<"&EDATE($A2,1))

	A	B	C	D	E	F	G
1	Месяц	Search выручка	Search расход	Search $ на $1	Social выручка	Social расход	Social $ на $1
2	Jan 2025	44,755	8,310	5.39	38,985	9,614	4.05
3	Feb 2025	45,949	8,676	5.30	42,677	10,746	3.97
4	Mar 2025	54,697	10,439	5.24	50,378	12,514	4.03
5	Apr 2025	56,835	10,977	5.18	41,105	10,502	3.91
6	May 2025	64,158	12,284	5.22	54,058	13,701	3.95
7	Jun 2025	58,377	11,282	5.17	48,726	12,325	3.95
8	Jul 2025	50,538	9,689	5.22	43,154	21,949	1.97
9	Aug 2025	50,739	9,621	5.27	49,856	25,112	1.99
10	Sep 2025	61,999	12,028	5.15	55,541	27,805	2.00
11	Oct 2025	69,634	13,300	5.24	54,669	27,070	2.02
12	Nov 2025	100,933	19,217	5.25	89,568	44,687	2.00
13	Dec 2025	117,557	21,957	5.35	100,727	50,387	2.00
14							
15							



DEMO 1 LIVE • 5 MINUTES

We do it live: from the file to a formula in a cell

1. **Upload** northstar_2025.xlsx to ChatGPT
2. **Prompt 1** — describe the file (slide 11)
3. **Prompt 2** — the “Ad return” sheet (slide 13)
4. **Download** the workbook and open it in Excel
5. **Click** an ad-return cell: a number → “replace the numbers with formulas”
6. **Out of time?** Show the ready frames from slide 14

What you should get: Search earns about $5 per ad dollar all year; Social drops to about $2 from July.

Have Claude or Copilot in Excel? The same prompt 2 in the side panel, no download.

Answer takes over 90 seconds — show the frame and move on.

“One number for the slide” — and the model picked the loudest

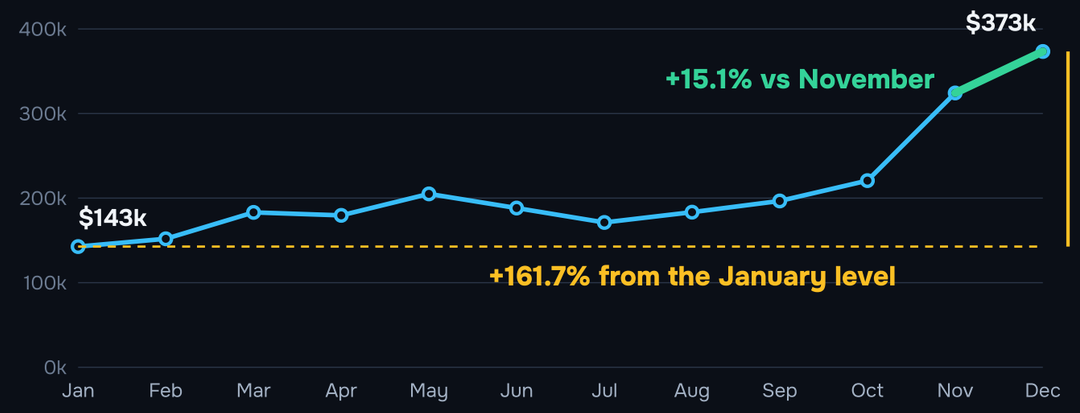
X WEAK “How much did sales grow this year? One number for the slide.”

Обработка заняла 1m 19s

+161,7% — рост выручки декабря к январю. northstar_2025(4)

+161.7% is December vs January. January is the weakest month, December is sales season. It sounds like growth; it’s the season.

Revenue by month



✓ GOOD “Compute growth three ways, give the base and period for each. Which number goes on the slide and why?”

первого». Ниже — три расчёта с явной базой сравнения.

Считаю продажи по **выручке** (`revenue`). Все периоды — **2025 год**. Формула: `(выручка периода / выручка базы − 1) × 100%`. northstar_2025(4)

Сравнение	База сравнения: период и выручка	Сравниваемый период и выручка	Рост
Декабрь к январю	Январь: **142 712,00**	Декабрь: **373 435,00**	**+161,7%**
Декабрь к ноябрю	Ноябрь: **324 344,50**	Декабрь: **373 435,00**	**+15,1%**
Второе полугодие к первому	Январь–июнь: **1 054 310,50**	Июль–декабрь: **1 469 658,00**	**+39,4%**

Почему на слайд — около 40%

Сравнение полугодий **охватывает весь год и меньше зависит от двух отдельных месяцев**. Для общего слайда об итогах года оно содержательнее:

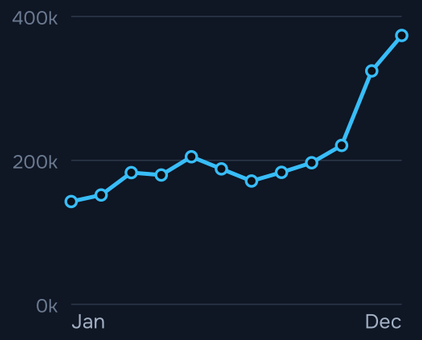
- **+161,7%** показывает, насколько декабрь выше января.
- **+15,1%** показывает динамику последнего мес
- **+39.4%** показывает разницу между двумя половинами года.

Three numbers with their base and a verdict: “the second half is about 40% above the first”.

Five questions, five charts — all from our file

How did it change over time?

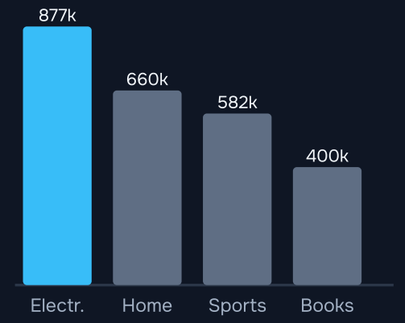
LINE



peak in Nov–Dec

Which is bigger?

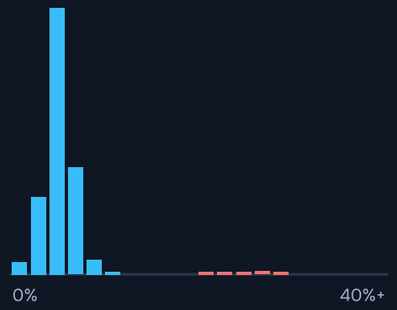
SORTED BARS



Electronics leads revenue

How are values spread?

HISTOGRAM



red tail: 22 rows with 14%+ returns

Are two numbers related?

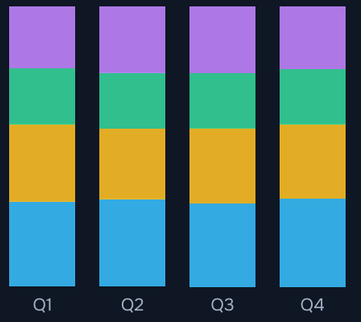
SCATTER



● Search ● Social Jan–Jun ● Social Jul–Dec: spend up, revenue not

What makes up the whole?

100% BARS



revenue shares by quarter: ● Search ● Social ● Email ● Organic

Ask like this: “Build one chart that answers …, label the axes and units.” Name the chart type if you know it; otherwise the model will suggest one.

BREAK

5 minutes

05:00

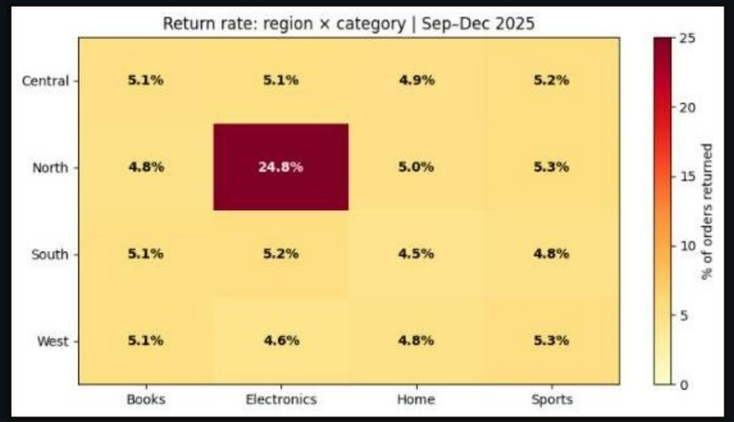
Next: a demo in a chat with a file — returns, reviews, a slide-ready chart. Then a large database, a one-page report and homework.

6% returns for the year. One cell shows 24.8%

Prompt 3 — the table

1	Compute the return rate	metric: returns ÷ orders
2	by region and category	breakdown: two axes
3	for September–December	the period Anna asked about
4	and show it as a heatmap.	type: the outlier shows as colour

RESULT: 24.8% IN ONE CELL



Prompt 4 — the text

1	For the warehouse with the highest return rate	the model finds the slice
2	sort the reviews into themes	text → categories
3	and compare with the other warehouses.	the base for comparison
4	Show 5 example reviews.	a check by eye

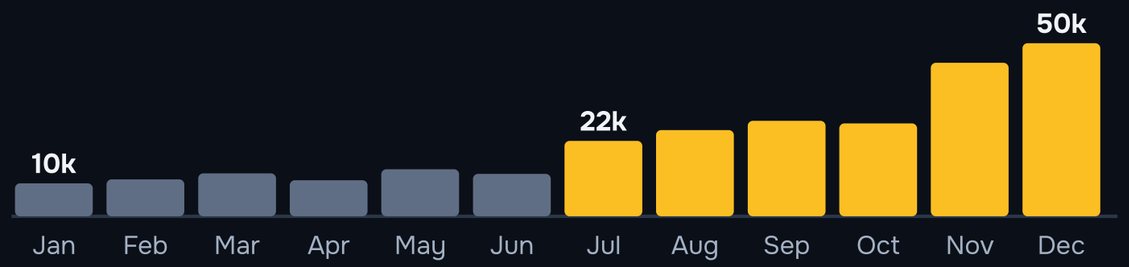
RESULT: N2 VS OTHER REVIEW THEMES · PROMPT 5 → SLIDE 21



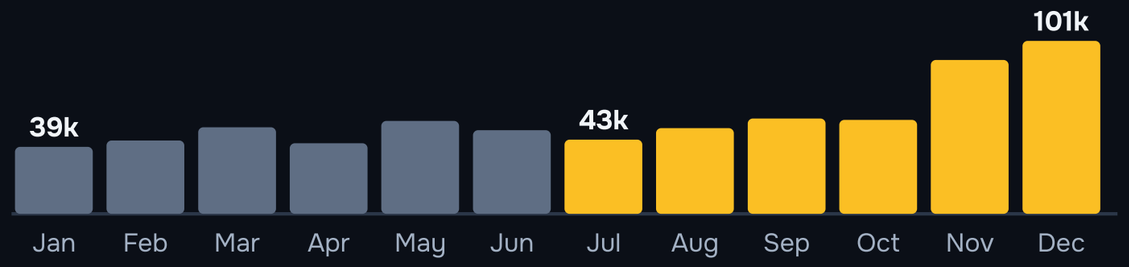
Two problems the annual total doesn’t show

1. Social: spend went up, revenue didn’t

Social ad spend by month, $



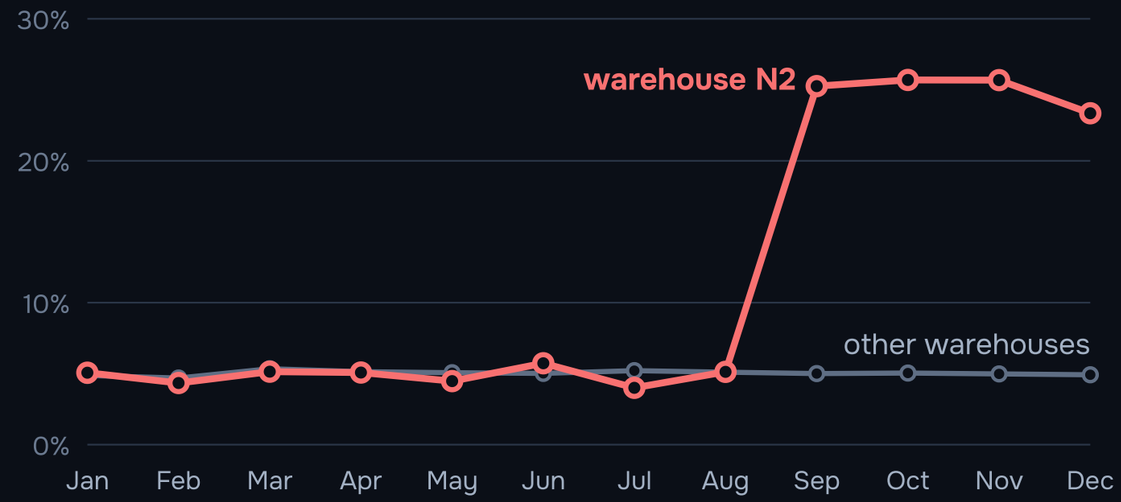
Social revenue by month, $



Second half vs first: spend **+184%**, revenue **+43%**. Revenue per $1 of ads: 4 → 2.

2. Warehouse N2: returns and damage complaints

Return rate by month, %

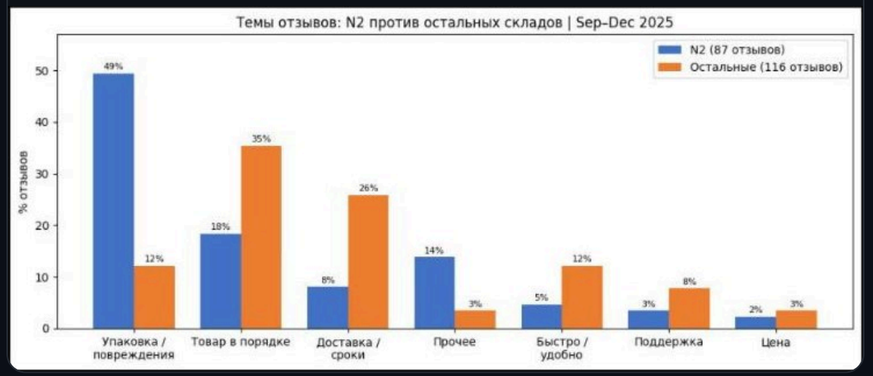


From September N2 returns are about 26%, others about 5%. Almost half of N2 reviews are about damaged parcels.

The number shows where; the text explains why. Reviews are data too.

You edit a chart with words, like text

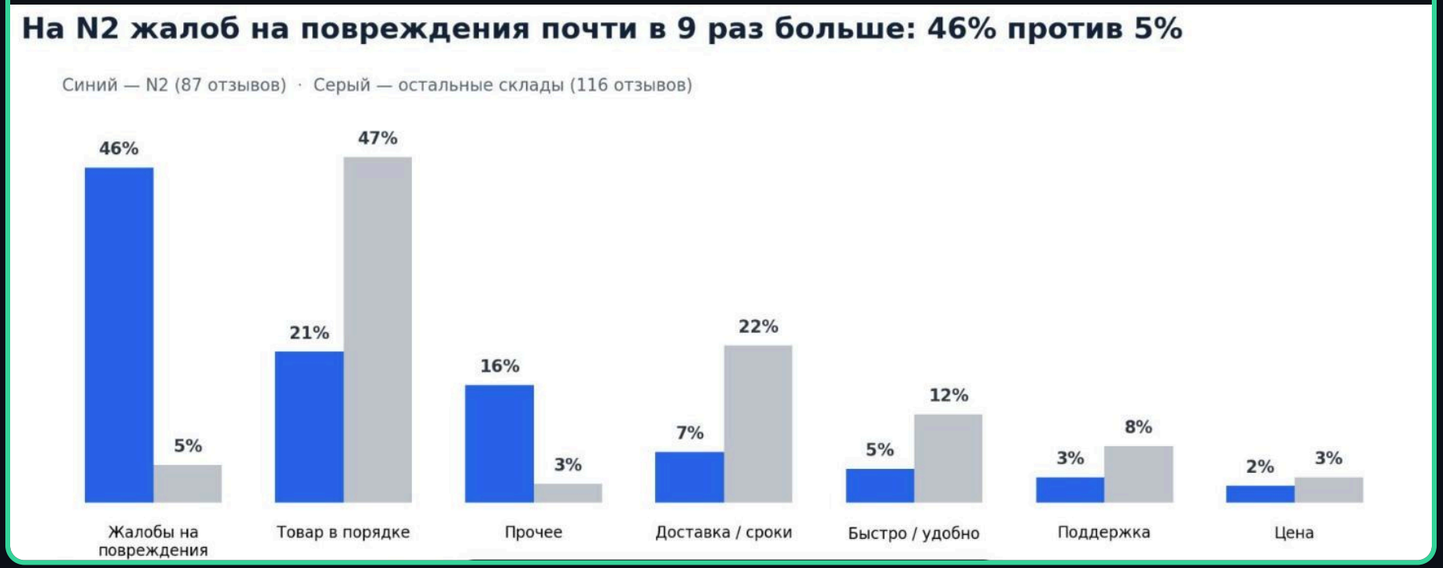
BEFORE: THE DEFAULT CHART



PROMPT

```
Rework this chart for a slide: the title
states the finding, bars sorted, value labels
on the bars, no legend and no gridlines, one
accent colour for N2, everything else grey.
Explain in 3 points what you changed.
```

AFTER: 36 SECONDS LATER



Title = finding. “Almost 9× more” instead of “Review themes”.

One colour. N2 blue, the rest grey — the eye finds the point at once.

Values on the bars, no legend or grid.

Three versions of a chart, three different numbers. Compare after every redo

Share of N2 reviews about damage, September–December

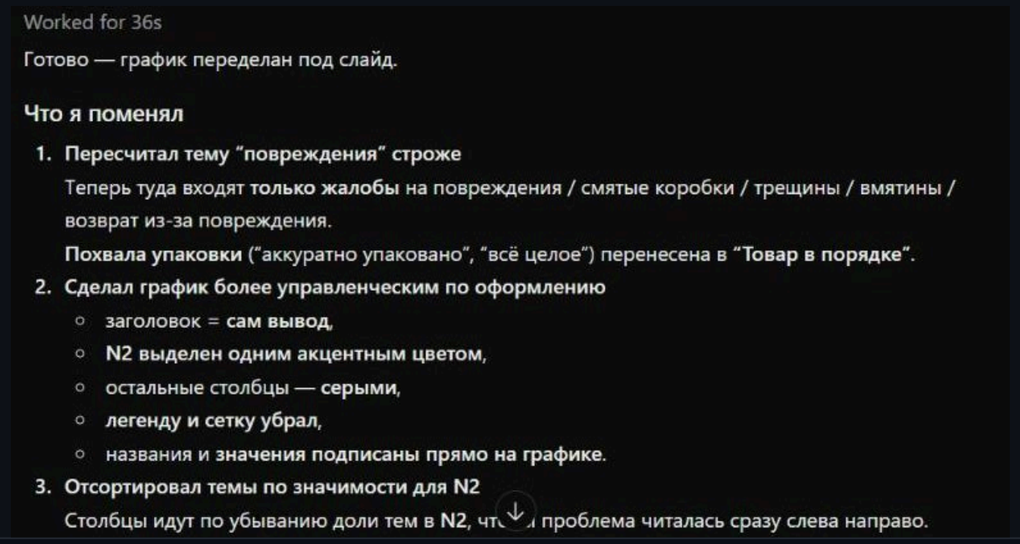
49 % first chart

46 % after restyling

44 % in the Word report

61 % by the labels the file was built from

THE MODEL EXPLAINS WHAT IT CHANGED



Worked for 36s

Готово — график переделан под слайд.

Что я поменял

1. **Пересчитал тему “повреждения” строже**
   Теперь туда входят **только жалобы** на повреждения / смятые коробки / трещины / вмятины / возврат из-за повреждения.
   Похвала упаковки (“аккуратно упаковано”, “всё целое“) перенесена в **“Товар в порядке“**.
2. **Сделал график более управленческим по оформлению**
   - заголовок = **сам вывод**,
   - **N2 выделен одним акцентным цветом**,
   - остальные столбцы — **серыми**,
   - **легенду и сетку убрал**,
   - **названия и значения подписаны прямо на графике**.
3. **Отсортировал темы по значимости для N2**
   Столбцы идут по убыванию доли тем в N2, чт… проблема читалась сразу слева направо.

1 Rows before and after

1,261 in the file → 1,260 after the duplicate is removed. How many were dropped, and why?

2 One point: against the source and the previous version

Find one number from the chart in the file and in the previous answer. Does it match?

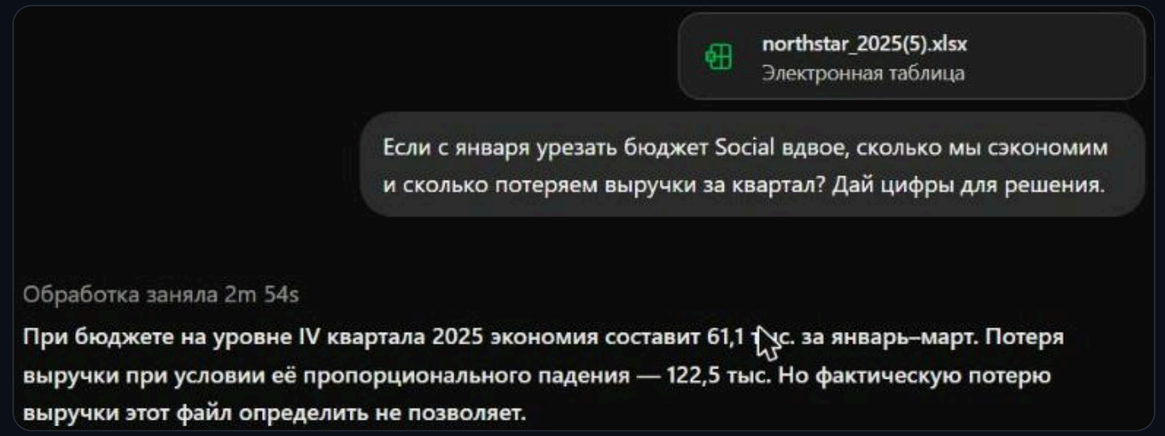
3 Percent of what

From which month, over what period? +161% and +15% are both right.

Check review themes by reading: open 10 reviews and count yourself.

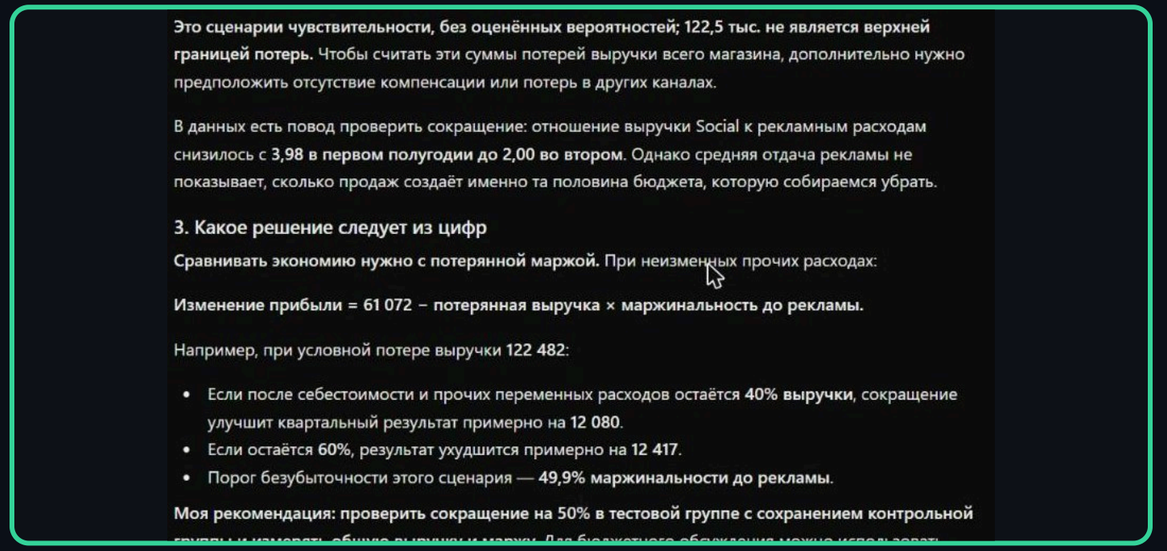
The first line is a scenario, not a forecast. Read the assumptions

QUICK The first line of the answer to “If we cut the Social budget in half from January, how much do we save and how much revenue do we lose per quarter?”



“Revenue loss — 122.5k” in the first line. That assumes revenue falls **together with** the budget.

✓ UNDER CONTROL The same answer, paragraphs below



The model itself says: the file can’t show the real loss; you need a test with a control group.

The channel database: three linked tables, updated daily

channels

26,544

rows · row = one channel

```
title
subscriber_count
country
language
```

videos

416,809

rows · row = one video

```
channel_id → channels
published_at
duration_seconds
tags
```

video_stat_observations

668,665

rows · row = one view measurement

```
video_id → videos
observed_at
views
```

channels → videos → video_stat_observations

An arrow is a link: a video holds its channel id, a measurement holds its video id. For the lesson: an anonymised sample of 1,943 channels and 46,132 videos.

SQL reads almost like a plain sentence

PROMPT

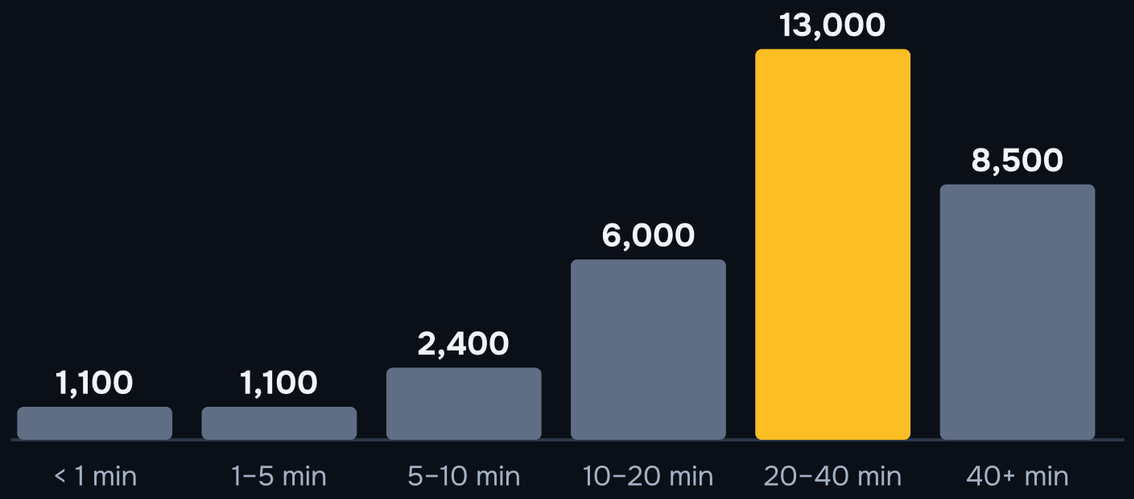
```
The database has a videos table: length and views of every video. Write an SQL query: which video length gets the most views?
Explain every line.
```

1	`SELECT duration_bucket,`	what to show: length
2	`COUNT(*) AS videos,`	how many videos in the group
3	`MEDIAN(views) AS typical`	typical views: half above, half below
4	`FROM videos`	from which table
5	`GROUP BY duration_bucket;`	what to group by

Describe the tables in the prompt. Without it the model guesses column names and the query fails.

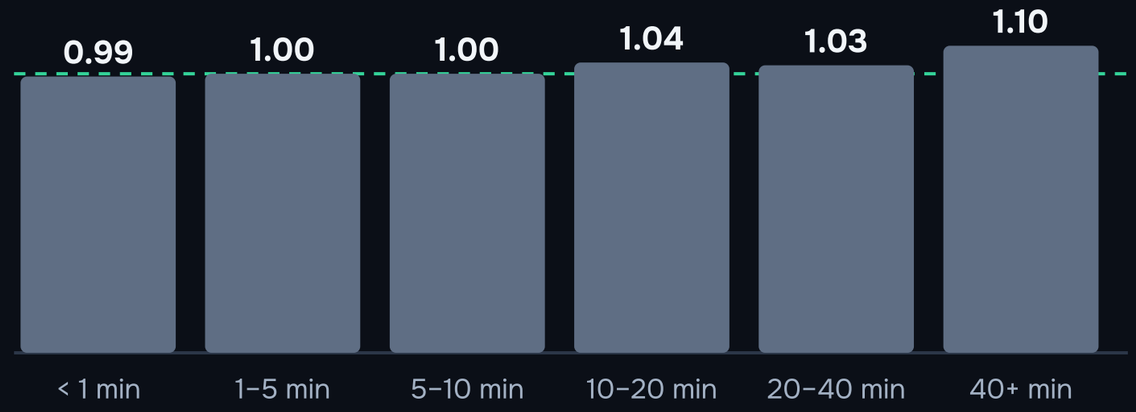
Long videos 12× more popular? No: big channels make them

First query: typical views by length



20–40 minutes gets 12× more than short videos? Compared with ordinary videos **of the same channel**, they are only 3% above usual. Big channels simply make long videos more often.

Second query: a video vs its own channel's usual video



1.0 means the same as the channel's usual video. Every length is about 1.0.

Look for an Export button first, then an API. Scraping is the last option

1. Export

An “Export to Excel / CSV” button exists almost everywhere: CRM, accounting, ad accounts, YouTube Studio, Google Analytics.

`fast · legal · no code`

2. Official API

```
{ "title": "Kitchen Basics",
  "subscriberCount": "184000",
  "videoCount": "412" }
```

A program asks the service and gets an answer like this. AI writes a script that turns it into table rows.

`this is how the slide 24 channel database was collected`

3. Scraping

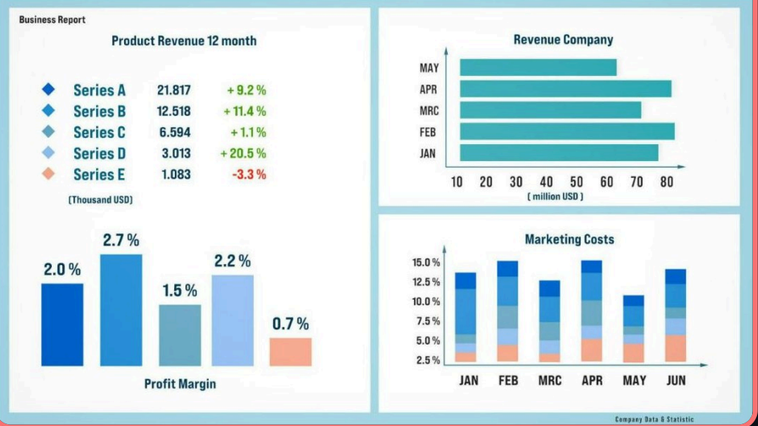
A program reads website pages like a person would. Before you do it, check:

1. the site’s terms allow it
2. no personal data
3. you don’t overload the site

Ask the AI: “What official export or API does system X have to download a year of sales?”

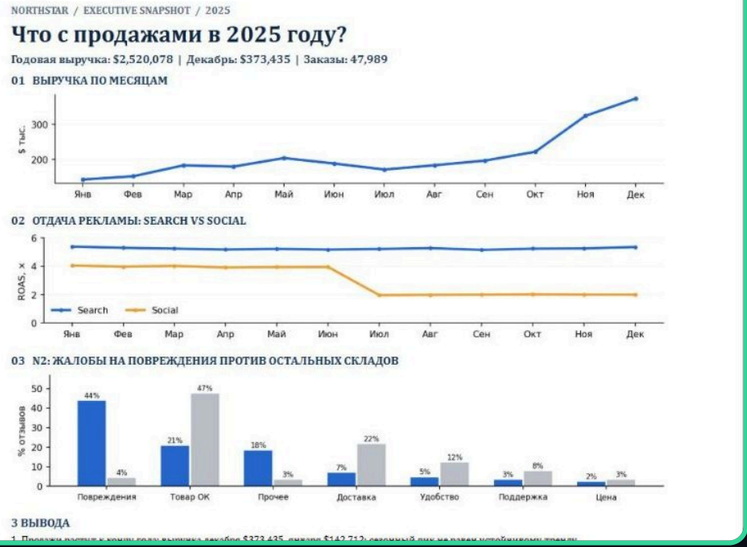
Send the manager the report file. The chat preview can lie

✗ PREVIEW IN THE CHAT



A placeholder picture: “Series A”, invented numbers. None of our analysis is here.

✓ THE DOWNLOADED .DOCX



The real file: the question, three charts, three findings, “what the data does not prove”.

PROMPT 6

```
Build a one-page report for a
manager in Word: the question, 3
charts (revenue by month, ad return,
N2 complaints), 3 findings and one
point on “what this data does not
prove”. And give me a short Python
script that builds the same 3 charts
from next month’s file.
```

RECIPE FOR NEXT MONTH

Bash

```
pip install pandas openpyxl matplotlib
python rebuild_northstar_charts.py next_month.xlsx
```

Результат появится в папке `charts`:

- `01_revenue.png` — выручка по месяцам.
- `02_roas.png` — отдача Search и Social.
- `03_reviews_N2.png` — темы отзывов N2 и остальных складов.

What you may upload to a chat

- **OK: practice, public, anonymised data**
  like Northstar; any service
- **Careful: internal company data**
  only in the work account (ChatGPT Business/Enterprise, Claude Team/Enterprise, Google Workspace) — data is not used for training there
- **Not without approval: customers, staff, finance, health**
  personal data — only in an approved environment with the data owner's consent

Before uploading, **delete unneeded columns**: names, phones, e-mails. Sales analysis doesn't need them.

In personal accounts, training on your data **depends on settings**: ChatGPT — "Improve the model for everyone", Gemini — "Keep Activity", Claude — only if you allowed it.

PART 5 • HOMEWORK

A pack you can send to a manager

File `northstar_2025.xlsx`. Answer Anna's e-mail: **how did sales go this year, and what should we check next?**

```
analysis_pack/
  analysis.xlsx    # AI formula + pivot + chart
  charts/          # 3 charts, one as before / after
  report.docx      # 1 page: question, 3 findings
  recipe.md        # prompt + script for January
  checks.md        # 60 seconds + "does not prove"
```

No paid AI in Excel needed: file to a chat → a workbook with formulas back → open in Excel.

Required: one statement that sounds plausible but the data does not prove.

Templates and rubric — in L08_05_Homework and demo_v2/student_starter.

SUMMARY

What to take away

1. ✓ **Describe a file**
   “describe the file, suggest 5 questions, change nothing”
2. ✓ **A formula in words**
   “Excel formulas, not pasted numbers” + click a cell
3. ✓ **A slide-ready chart**
   one question, one chart, title = finding
4. ✓ **A one-page report**
   open the downloaded file, attach the recipe
5. ✓ **A one-minute check**
   rows, one point, percent of what

Next lesson — L09: the Claude ecosystem and artifacts. The same analysis becomes an interactive page opened by a link.